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A EXPERIMENTAL DETAILS

In this section, we show more details about the experiments we conducted in the paper.

A.1 DATASETS AND BASE MODELS

We select instruction-following demonstrations for the base LLMs to learn in context from open-sourced instruction fine-tuning datasets: (a) SkillMix-4k (Kaur et al., 2024) is an automated approach for synthesizing diverse, high-quality IFT data. It primarily involves two stages: first leveraging the LLMs to propose a set of critical "skills" for instruction-following, from which a pair of skills are randomly chosen to facilitate synthetic data generation based on powerful LLMs. We show two examples from SkillMix in Fig. 6 and Fig. 7. (b) Evol-Instruct-70k (Xu et al., 2024) contains 70k training examples with varying complexity and is well-known for the use to build the series of WizardLM models. SkillMix-4k is the primary dataset for all experiments in the paper and Evol-Instruct-70k is solely used for scaling experiments and ablation study of data quality.

We compare ICL performance across multiple models that have sufficiently large context windows:

1. Mistral-7B-v0.2 (Jiang et al., 2023) has 7.3 billion model parameters. On many established benchmarks, it outperforms LLMs that have significantly more parameters, such as Llama-2-13B and Llama-1-34B. The trained context length of Mistral-7B-v0.2 is 32k tokens.
2. Llama-3.1-8B (Dubey et al., 2024) is the newest and most powerful 8B model from Meta when we are writing, and it supports multilingual dialogue use cases. It supports input texts containing up to 128k tokens.

In some experiments, the following base LLMs are used:

1. Llama-2-7B-80k (Fu et al., 2024) is a variant of Llama-2-7B model fine-tuned with 80k context on a carefully designed long-document data mixture.
2. Mixtral-8x22B-v0.1-4bit is a variant of Mixtral-8x22B-v0.1 (Jiang et al., 2024), which is a pre-trained generative sparse Mixture of Experts, quantized with 4-bit precision. It contains ~ 176 B parameters and ~ 44 B active during inference, and it has a 65k context window.

We use the same decoding configuration as what is used in URIAL (Lin et al., 2024). Concretely, we employ greedy decoding, i.e., temperature = 0.0, for all models, including base and instruction fine-tuned models, to maximize reproducibility and secure a fair and robust evaluation. Top-p = 1.0 is adopted to keep the full cumulative probability distribution. Besides, we use repetition penalty = 1.15² on base models to prevent degeneration.

A.2 EVALUATION

MT-Bench (Zheng et al., 2023), consists of 80 high-quality and challenging questions that have two-round interaction, designed to examine the multi-turn conversation and instruction-following capability of models. It features 8 common categories of user prompts: coding, math, reasoning, extraction, roleplay, writing, humanities/social science, and STEM.

AlpacaEval 2.0 (Li et al., 2023a) provides 805 test instructions, on which we generate new responses using the target model, and then calculate the score by competing with the baseline model (i.e., GPT-4-Turbo) judged by a designated automatic evaluator. We apply the AlpacaEval 2.0 benchmark in our experiments to ensure that the effective demonstrations found through greedy search don't overfit to MT-Bench.

A.3 GREEDY SEARCH

Firstly, we randomly sample 100 examples from the high-quality IFT dataset, SkillMix-4k, and then create 100 new 4-shot prompts by adding each one of the 100 sampled examples, as the fourth

²As in the codebase at https://github.com/Re-Align/URIAL/blob/main/run_scripts/mt-bench/_run_mt_bench.sh.

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### Skills: developing training programs, digital proficiency

### Instruction
I'm planning to develop a digital literacy program specifically for seniors at our community center. Could you guide me on the essential components to include, how I can leverage digital tools effectively, and provide some concrete examples or success strategies from similar programs?

### Response
Creating a tailored digital literacy training program for seniors is key to enhancing their technology engagement. Below is a comprehensive guide featuring improved content based on earlier feedback:

1. Assessing Learning Needs:
Conduct surveys or focus groups to understand seniors' technological backgrounds and learning objectives. This data will guide the customization of training content.
2. Structuring the Program:
Organize the program into manageable modules:
- Basic Computing Skills: Instructions on fundamental operations of devices, incorporating user-friendly technology like Chromebooks, which are simpler for beginners.
- Internet Navigation: Teach safe browsing, using straightforward and secure websites for everyday tasks.
- Email and Communication Tools: Demonstrate setting up and using email and video calling, with adjustments for larger display settings to aid visibility.
- Online Safety: Educate on creating strong passwords with tools like LastPass, identifying secure websites, and recognizing scams.
3. Using Digital Tools in Training:
Integrate specific digital tools to enhance learning:
- Interactive Tutorials: Implement platforms like GCGlobal that offer senior-friendly interactive lessons.
- Feedback Mechanisms: Use simple, engaging quizzes through Kahoot! to review and reinforce learning.
4. Scheduling and Accessibility:
Plan short, regular sessions. Ensure the learning environment is comfortable and accessible, considering both physical and online settings.
5. Facilitating Engagement:
- Peer Learning: Support seniors working in pairs to promote mutual assistance and engagement.
- Ongoing Support: Establish a support system such as a helpline or dedicated email for ongoing help, coupled with monthly follow-up sessions.
6. Case Studies and Success Stories:
Refer to successful programs like AARP's TEK workshops, highlighting their positive impact on senior confidence with technology.
7. Measure Impact and Adapt:
Use analytics tools to monitor participation and feedback, continuously refining the program based on empirical data.

By focusing on these enhanced, specific steps and maintaining a dynamic feedback loop, your digital literacy program will significantly improve seniors' technological fluency and confidence.

### Query Type: Planning
### Data Generator: gpt-4-turbo-2024-04-09
```

Figure 6: An example of SkillMix describes a conversation about planning and includes two skills, developing training programs and digital proficiency.

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### Skills: performance metrics analysis, writing review articles

### Instruction
I am working on a review article about various performance metrics used in machine learning models for a class project. Specifically, I want to understand the real-world implications and methodological differences between the F1 Score and AUC-ROC. Can you explain and compare these metrics based on the following excerpt from a research paper?

Excerpt:
"In evaluating classifier performance, the F1 Score and AUC-ROC are pivotal metrics. The F1 Score, a harmonic mean of precision and recall, crucially balances the trade-offs between false positives and false negatives. On the other hand, AUC-ROC, which measures the area under the receiver operating characteristic curve, provides an aggregate measure of performance across all classification thresholds. Notably, AUC-ROC is beneficial for evaluating classifiers on imbalanced datasets as it is unaffected by the class distribution."

### Response
Interpretation and Analysis of F1 Score and AUC-ROC

The provided excerpt discusses two crucial performance metrics for classifiers in machine learning: the F1 Score and AUC-ROC. Each has distinct utilities and provides unique insights into a model's performance under different conditions.

F1 Score: Balancing Precision and Recall
The F1 Score is the harmonic mean of precision and recall, ensuring that both false positives and false negatives are equally weighted. This metric is essential in situations where both types of errors are costly. For instance, in fraud detection systems, it is critical to minimize both false alarms (false positives) and missed frauds (false negatives). The F1 Score is optimal here as it balances sensitivity (recall) with the precision needed to avoid excessive false alerts.

AUC-ROC: Evaluating Performance Across Thresholds
AUC-ROC measures the area under the receiver operating characteristic curve, providing a comprehensive view of model performance across all classification thresholds. This metric is especially valuable in imbalanced datasets, common in medical diagnostics, where the prevalence of one class significantly outweighs the other. AUC-ROC assists in understanding how well a model can distinguish between classes, independent of the class distribution.

Comparison and Practical Application
While the F1 Score gives a precise measure of model accuracy at a specific threshold, focusing on the balance between precision and recall, AUC-ROC offers insight into the model's ability to discriminate between classes across thresholds. This makes AUC-ROC suitable for preliminary model analysis and scenarios where class imbalance might skew the perception of model effectiveness provided by other metrics like accuracy.

Conclusion
Choosing between F1 Score and AUC-ROC, or using them in conjunction, depends on the specific needs of the classification task and the cost implications of errors. Understanding these metrics' implications can significantly enhance the quality and depth of your review article, providing a clearer guide for model evaluation in diverse practical scenarios. Both metrics together provide a robust framework for evaluating classifier performance, especially in complex real-world settings.

### Query Type: Interpretive Explanation
### Data Generator: gpt-4-turbo-2024-04-09
```

Figure 7: An example of SkillMix describes a conversation about interpretive explanation and includes two skills, performance metrics analysis and writing review articles.